

Take-Home Assignment: Expense Management System with Multi-Level Approvals & Role Management

Overview

Design a scalable enterprise-grade Expense Management Platform that supports configurable approval workflows and role-based access management.

This assignment is intentionally open-ended. We expect you to make reasonable assumptions, define product behavior, and explain technical and architectural decisions clearly.

The goal is to evaluate:

- Product thinking
 - System design skills
 - Technical depth
 - Scalability awareness
 - Data modeling
 - Workflow design
 - Security and permissions understanding
 - Tradeoff analysis
 - Communication clarity
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Problem Statement

Your company is building a SaaS-based expense management platform used by organizations of different sizes — from startups to large global enterprises.

Employees can submit expenses for reimbursement or direct company payment.

The platform should support:

- Expense creation and tracking
- Multi-level approval workflows
- Role-based access management

Your Task

Design the system end-to-end.

You are expected to define:

- Functional requirements
- Non-functional requirements
- APIs
- Data models
- System architecture
- Workflow engine approach
- Role/permission model
- Scalability strategy
- Security considerations
- Failure handling
- Audit mechanisms

You may make assumptions where needed, but clearly document them.